

Ian Cheung

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EDUCATION

University of California, Berkeley

Expected Graduation: May 2028

Applied Mathematics and Data Science, B.A.

GPA: 3.88

Relevant Coursework: Structure and Interpretation of Computer Programs, Foundations of Data Science, Multivariable Calculus, Discrete Mathematics, Linear Algebra, Data Structures, Probability & Statistics, Machine Learning, Data Engineering

EXPERIENCE

Bridging Berkeley *Mathematics Mentor, Longfellow Middle School*

September 2025 – Present

- Provide individualized math support to **50 middle school students** on a weekly basis.
- Communicate and break down mathematical concepts to different classes into easily digestible lessons.

OpenBrackets *Vice President of Education · Python Instructor*

April 2023 – April 2025

- Delivered **8-week** remote Python programming course for **4** semesters to **100+ middle school students** with no prior coding exposure.
- Designed interactive Google Colab notebooks covering programming fundamentals, data structures, logic, and basic algorithms.

PROJECTS

Fischl (*Creator & Lead Developer*) *Discord.py · Flask · Firebase · PostgreSQL · AI*

February 2022 – Present

- Developed and maintain a modular Discord bot with ticket system & automated queues, serving **300,000+ users** with 99.9% uptime.
- Engineered a custom **Agentic Ticket System** using Llama-based models on **Groq** for high-speed inference with a custom **multi-step tool-calling orchestration layer** to autonomously classify intent and execute administrative operations.
- Built full-stack **web dashboard** with Flask backend, enabling server admins to view AI analytics and configure server settings.
- Deployed monitoring infrastructure with Prometheus and Grafana to track performance metrics and ensure reliability.

Analysis of California Traffic Collisions *Python · Pandas · NumPy · Scikit-learn · Matplotlib*

August 2024

- Analyzed **5 million** collision records using **data cleaning, lineage, provenance**, EDA, feature engineering, and model deployment.
- Trained and tuned **random forest** and **logistic regression** models to identify key factors of collision severity.
- Developed **actionable insights** and data-driven policy recommendations with visualizations for roadway safety improvements.

RecycleSorter *HTML · CSS · JavaScript · TensorFlow*

January 2023

- Built and deployed image processing pipeline with **TensorFlow.js** to distinguish recyclables, compost, or landfill items.
- Awarded **Most Creative Award** at GunnHacks 9.0 for the AI-powered web app to **classify waste** from live camera footage.

GunnHacks 11.0 (*Web Developer & Workshop Host*) *React · Vercel*

November 2024 – January 2025

- Deployed and maintained official website annual high school hackathon, streamlining logistics and registration for **40+ participants**.
- Hosted **introductory workshop** on web development fundamentals, teaching HTML, CSS, and JavaScript to beginner hackers.

Librarian Attendance System (*Chief IT Developer*) *HTML · CSS · JavaScript*

September 2021 – May 2022

- Created **automated clock-in system** for La Salle College librarians, reducing manual spreadsheet data entry time by **90%**.
- Designed **web interface** enabling quick check-in/out using student ID card with timestamp logging and weekly shift report.

TECHNICAL SKILLS

AI & Machine Learning: Llama, GPT, Claude, Gemini, Groq (LLaMA), RAG, LangChain, Tool-Calling, Prompt Engineering

Data Science & Analysis: Python (Polars, Pandas, NumPy, Scikit-learn), R, Data Structures, EDA, Statistical Modeling

Infrastructure & Databases: PostgreSQL, MariaDB, Cloudflare, Firebase, Docker, Prometheus, Grafana, Linux, Git, Nginx, Vercel

Software Development: Flask, FastAPI, Node.js, React, JavaScript, TypeScript, Java, Discord.py